



Reactivity Series & Corrosion of Metals

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Reactivity series

What is the reactivity series of metals?

- The chemistry of the metals is studied by analysing their reactions with water and acids
- Based on these reactions a reactivity series of metals can be produced
- The series can be used to place a group of metals in **order of reactivity** based on the observations of their reactions with water and acids
- The non-metals hydrogen and carbon are also included in the reactivity series as they are used to extract metals from their oxides

Reactivity series order

- The order of the reactivity series is:
 - Potassium
 - Sodium
 - Calcium
 - Magnesium
 - Aluminium
 - **Carbon (non-metal)**
 - Zinc
 - Iron
 - **Hydrogen (non-metal)**
 - Copper
 - Silver
 - Gold

How to remember the reactivity series

- The order of this reactivity series can be memorised using the following mnemonic
 - "Please send cats, monkeys and cute zebras into hot countries signed Gordon"

Reactivity Series Mnemonic

Metal	Abbreviation
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Your notes

Most reactive	
Potassium	P - Please
Sodium	S - send
Calcium	C - cats,
Magnesium	M - monkeys,
Aluminium	A - and
Carbon	C - cute
Zinc	Z - zebras
Iron	I - into
Hydrogen	H - hot
Copper	C - countries
Silver	S - signed
Gold	G - Gordan
Least reactive	

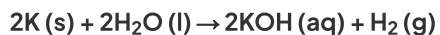
Reaction of metals with cold water

- Potassium, sodium and calcium react with cold water to form a metal hydroxide and hydrogen gas:



- Potassium

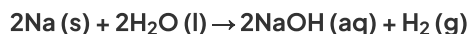
- Reacts violently with cold water



- Sodium

- Reacts quickly with cold water





- Calcium
 - Reacts less strongly with cold water

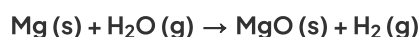
calcium + water → calcium hydroxide + hydrogen



Reactions of metals with steam

- Magnesium does not react much with cold water but reacts with steam to form a metal oxide and hydrogen gas:

magnesium + steam → magnesium oxide + hydrogen



Reaction with dilute hydrochloric acid

- Metals **above hydrogen** in the reactivity series react with dilute hydrochloric acid
 - The more reactive the metal then the more vigorous the reaction will be
 - Metals that are placed high on the reactivity series such as potassium and sodium are very dangerous and react **explosively** with acids
- Metals **below hydrogen** in the reactivity series do not react with dilute hydrochloric acid
- When hydrochloric acid reacts with a metal it forms a **salt** and **hydrogen gas**:

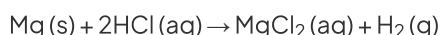
metal + acid → salt + hydrogen

Metals reacting with dilute hydrochloric acid

- Metals that do react with dilute hydrochloric acid include:

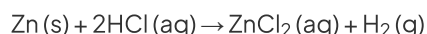
1. Magnesium

magnesium + hydrochloric acid → magnesium chloride + hydrogen



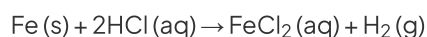
2. Zinc

zinc + hydrochloric acid → zinc chloride + hydrogen



3. Iron

iron + hydrochloric acid → iron(II) chloride + hydrogen



Metals that do not react with dilute hydrochloric acid

- Metals that do **not** react with dilute hydrochloric acid include:



Your notes



Your notes

1. Copper

copper + dilute hydrochloric acid $\rightarrow$ no reaction

2. Silver

silver + dilute hydrochloric acid $\rightarrow$ no reaction

3. Gold

gold + dilute hydrochloric acid $\rightarrow$ no reaction

- The more reactive the metal, the more vigorous the reaction with dilute hydrochloric acid is

Deducing the order of reactivity

- The order of reactivity of metals can be deduced by making experimental observations
 - The more vigorous the reaction, the more reactive the metal
- If two metals both react with water, the more vigorous reaction shows the more reactive metal
- If metals do not react with water, their order can be deduced from their reactions with dilute hydrochloric acid
- Temperature change can also be used
 - The greater the temperature change in a reaction involving a metal, the more reactive the metal is



Explaining reactivity

Extended tier only

- Metal atoms form **positive ions** by loss of electrons when they react with other substances
- The **tendency** of a metal to lose electrons is a measure of how reactive the metal is
- A metal that is high up on the series loses electrons easily and is thus **more reactive** than one which is lower down on the series

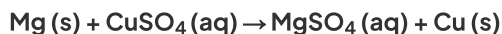
Displacement reactions between metals and aqueous solutions of metal salts

- Any metal will displace another metal that is **below** it in the reactivity series from a solution of one of its salts
- This is because more reactive metals lose electrons and form ions more readily than less reactive metals, making them better **reducing agents**
- The less reactive metal is a better electron acceptor than the more reactive metal, thus the less reactive metal is reduced
 - OILRIG: reduction is gain of electrons

Magnesium + copper sulfate

- Magnesium is a reactive metal and can displace copper from a copper sulfate solution
- Magnesium loses its electrons more easily and the ion of the less reactive metal, copper, will gain these electrons to form elemental copper
- This is easily seen as the more reactive metal slowly **disappears** from the solution, **displacing** the less reactive metal

magnesium + copper sulfate → magnesium sulfate + copper



- The blue colour of the CuSO_4 solution **fades** as colourless magnesium sulfate solution is formed
- Copper coats the surface of the magnesium and also forms solid metal which falls to the bottom of the beaker

The reaction between magnesium and copper sulfate



Your notes

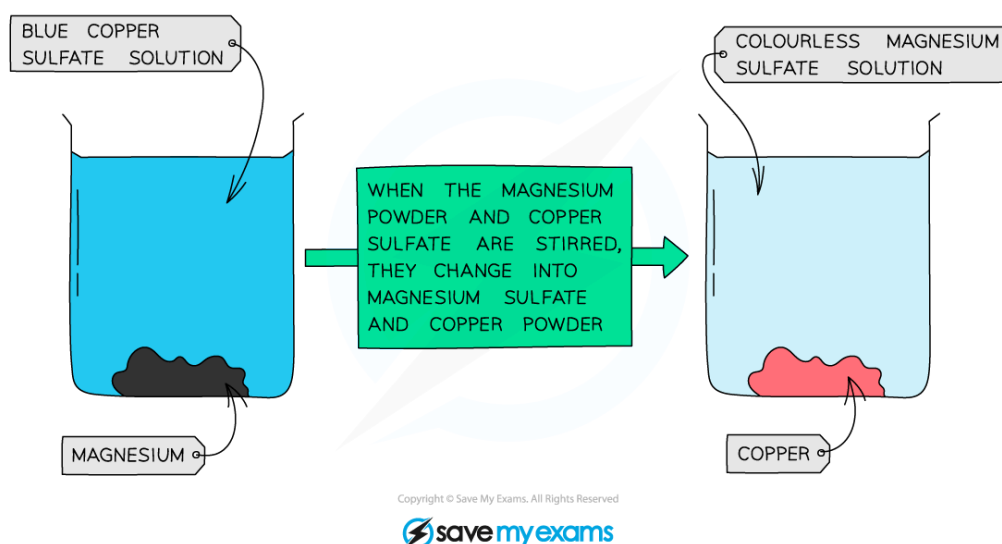


Diagram showing the colour change when magnesium displaces copper from copper sulfate

- By combining different metals and metal salts solutions it is possible to come up with a relative reactivity order

Metal Solutions Displacement Table

Mixture	Products	Equation for Reaction
Magnesium and iron(II) sulfate	Magnesium sulfate and iron	$\text{Mg} + \text{FeSO}_4 \rightarrow \text{MgSO}_4 + \text{Fe}$
Zinc and iron chloride	Zinc chloride and iron	$\text{Zn} + \text{FeCl}_2 \rightarrow \text{ZnCl}_2 + \text{Fe}$
Iron and silver nitrate	Iron(II) nitrate and silver	$\text{Fe} + \text{AgNO}_3 \rightarrow \text{Fe}(\text{NO}_3)_2 + 2\text{Ag}$
Copper and iron(II) chloride	No reaction	-----
Silver and copper(II) sulfate	No reaction	-----
Zinc and magnesium chloride	No reaction	-----

- From this table we can deduce the order of reactivity:
 - Magnesium and zinc are both more reactive than iron but magnesium is more reactive than zinc



Your notes

- Copper and silver are both less reactive than iron but silver is less reactive than copper
- The order of reactivity of the metals tested can be therefore be deduced as:
 - $\text{Mg} > \text{Zn} > \text{Fe} > \text{Cu} > \text{Ag}$

Reactivity of aluminium

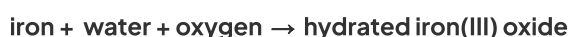
- Aluminium is high in the reactivity series, but in reality, it does not react with water and the reaction with dilute acids can be quite slow
- This is because it reacts readily with oxygen, forming a protective layer of aluminium oxide which is very thin
- This layer prevents reaction with water and dilute acids, so aluminium can behave as if it is unreactive



Rusting of iron

What is rusting?

- Rusting is a chemical reaction between iron, water and oxygen
 - Rust is the reddish-brown compound product called hydrated iron(III) oxide
- Both oxygen and water must be present for rusting to occur
 - Steel, which contains iron, also rusts under the same conditions
- During rusting, iron is oxidised



Investigating rusting

- To investigate the conditions required for rusting, prepare three test tubes as shown in the diagram
- The oil layer in the 2nd tube keeps out air and the water has been boiled so that no air is left in it
- The calcium chloride in the 3rd tube is used to remove any moisture in the air
- After a few days, the iron nail in the 1st tube will be the only nail to show signs of rust

Requirements for rusting

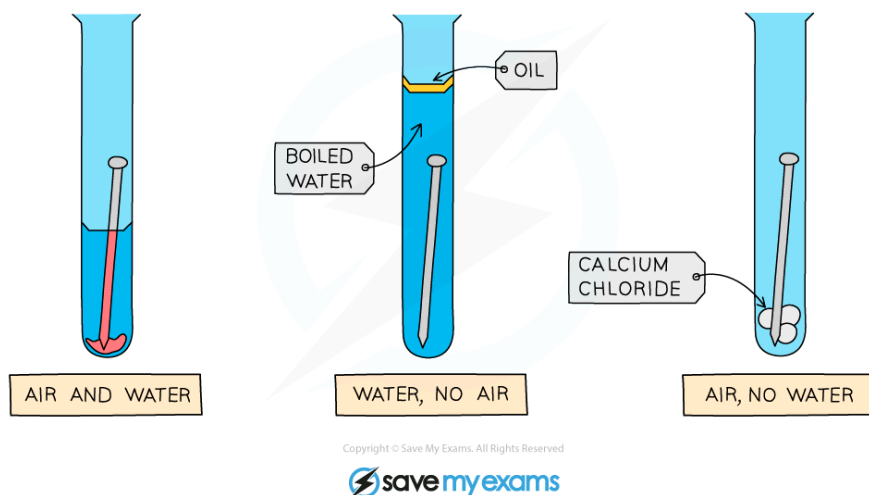


Diagram showing the requirements of oxygen and water for rust to occur: only the nail exposed to both air and water rusts

Rust prevention methods

- Rust can be prevented by coating iron or steel with barriers that prevent the iron from coming into contact with water and oxygen
- However, if the coatings are removed or scratched, the iron or steel is once again exposed to water and oxygen and will rust

Table to show common barrier methods

Common barrier methods	
Grease	Oil
Paint	Plastic



Examiner Tips and Tricks

Only iron or steel (an alloy made from iron) can **rust**. If any other metal oxidises in air causing the metal to break down, you should say that the metal has **corroded**.



Your notes



Galvanising & sacrificial protection

Extended tier only

Sacrificial Protection

- Iron can be prevented from rusting using the **reactivity series**
- A **more** reactive metal can be attached to a **less** reactive metal
- The more reactive metal will oxidise and therefore corrode first, protecting the less reactive metal from corrosion

Zinc bars on the side of steel ships

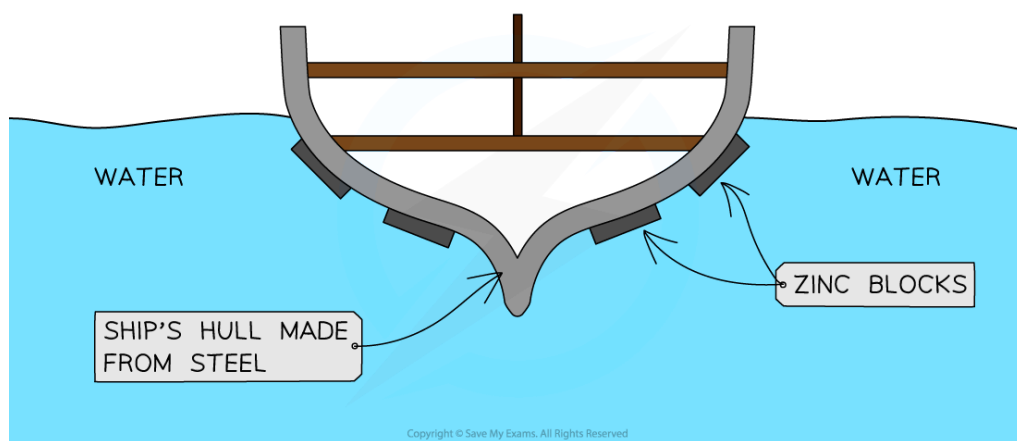
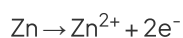


Diagram to show the use of zinc bars on the sides of steel ships as a method of sacrificial protection

- Zinc is more reactive than iron therefore will lose its electrons more easily than iron and is oxidised more easily:



- The iron is less reactive therefore will not lose its electrons as easily so it is not oxidised; the zinc is **sacrificed** to protect the steel
 - The zinc blocks do not need to cover the whole hull
 - As long as they are electrically connected to the steel, they can protect nearby areas
 - They are often placed on edges and joints where corrosion is most likely.
- For continued protection, the zinc bars have to be replaced before they completely corrode

Galvanising



Your notes

- **Galvanising** is a process where the iron to be protected is coated with a layer of zinc
- This can be done by electroplating or dipping it into molten zinc
- ZnCO_3 is formed when zinc reacts with oxygen and carbon dioxide in the air and protects the iron by the barrier method
- If the coating is damaged or scratched, the iron is still protected from rusting by **sacrificial protection**



Examiner Tips and Tricks

You may be asked to explain why a metal is/is not suitable as a method of preventing an iron/steel object from rusting. Remember that if it is higher in the reactivity series than iron, it will be suitable for sacrificial protection as it will be oxidised instead of iron. If it is lower in the reactivity series than iron, it would not be suitable as iron would be oxidised, causing it to rust.